

Combatting Synthetic Misrepresentation: A Deep Learning Approach to Detect AI Images in Etsy Listings

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Abstract—The integrity of the marketplace at Etsy is anchored on the authenticity of the products, which now is under threat due to the spread of hyper-realistic AI-generated images. The lines between handmade products and artificial representations have become increasingly unclear with the introduction of advanced generative models, causing a tangible trust crisis among consumers. This paper addresses the detection of these synthetic images through a fine-tuned MobileNetV2 architecture, specifically optimized for resource-constrained but high-precision settings. By increasing the input resolution to 224x224 and strategically unfreezing deeper convolutional layers, our network successfully isolates high-frequency artifacts and textural anomalies that remain invisible to standard baseline models. We further refine the decision boundary by conducting a rigorous manual grid search, intentionally varying the classification threshold to counteract the model’s natural bias toward over-predicting synthetic content. Our approach achieves a peak F1-score of 0.7719 at a conservative threshold of 0.65. This study underscores the paramount significance of spatial resolution and decision calibration in the ongoing arms race against contemporary generative content, offering a practical framework for platform moderation.

I. INTRODUCTION

Etsy operates as a large, internationally recognized marketplace that is fundamentally distinct from traditional e-commerce giants like Amazon or AliExpress. Its brand identity is heavily built on the promise that all products are singular, custom-made, hand-crafted, or vintage. Users navigate to the site specifically anticipating an interaction with human creators, expecting to purchase physical, thoughtfully crafted goods.

However, the e-commerce landscape has experienced a seismic shift with the abrupt and dramatic rise of highly developed Generative AI (GenAI) platforms. Tools such as Midjourney, OpenAI’s DALL-E 3, and Stable Diffusion have democratized the creation of studio-quality imagery. Consequently, a massive “Trust Crisis” has begun to permeate the Etsy ecosystem. Bad-faith vendors are now entirely capable of generating aesthetically perfect, hyper-realistic images of products that do not actually exist in the physical world.

These fake listings are highly appealing to unsuspecting buyers. The inevitable result, however, is the shipment of either

no goods at all or low-quality, mass-produced approximations of the AI image. This leads to severe customer dissatisfaction, chargebacks, and a profound loss of platform credibility. When a consumer can no longer inherently trust the visual representation of a listing, the core philosophy underpinning the marketplace is critically undermined.

This project directly tackles this growing problem by developing a high-precision binary classifier designed to distinguish between Authentic Listing Images (Label 0) and AI-Generated Images (Label 1). Our module professor provided us with a challenging, bounded task containing a hidden, held-out test set. This constraint required that our model do more than simply memorize training data; it had to generalize robustly to completely unknown, real-world data distributions.

When we first approached the dataset, initial experimentation revealed that a basic Random Forest classifier, trained on significantly reduced 32x32 pixel images, was practically useless. It operated only marginally above random guessing. It became immediately clear that modern AI detection requires a fundamentally different approach. We pivoted our focus toward identifying high-frequency textural artifacts—the microscopic, mathematical anomalies left behind by the generative diffusion process. We hypothesized that only Deep Convolutional Neural Networks (CNNs) processing high-resolution inputs could effectively isolate and analyze these micro-patterns. In this paper, we detail our architectural decisions, our iterative training approach, and the critical post-processing calibrations necessary to ensure real-world viability.

II. RELATED WORK

The detection of synthetic imagery is a highly dynamic and rapidly evolving subfield of digital forensics. As generative models have grown in sophistication, so too have the methodologies required to detect them, resulting in a continuous technological arms race.

A. Early Generative Adversarial Networks (GANs)

In the earlier days of deep learning, detecting synthetic images was a relatively straightforward task. Initial Generative Adversarial Networks (GANs), such as DCGAN or the early

iterations of StyleGAN, consistently produced recognizable, mathematically deterministic fingerprints in their visual output. These were most visibly manifested as "checkerboard artifacts." These anomalies occurred when receptive fields overlapped unevenly during the upsampling (transposed convolution) layers of the generator network. By employing simple frequency domain analyses, such as the Discrete Cosine Transform (DCT), forensic researchers could easily isolate and identify these artifacts without requiring massive neural networks.

B. The Age of Diffusion Models

The current state of generative architecture has shifted almost entirely away from GANs in favor of Diffusion models. These models operate on a fundamentally different paradigm, synthesizing images by iteratively reversing a Gaussian noise process. Because of this noise-denoising pathway, the upsampling artifacts characteristic of GANs are practically non-existent. Modern diffusion models generate remarkably smooth gradients, highly coherent global lighting, and nearly perfect macro-structures. Visually, they are often indistinguishable from professional photography, forcing forensic researchers to look deeper than surface-level anomalies.

C. Existing Forensic Practices

Contemporary forensic studies generally fall into one of two primary categories:

Statistical Noise Analysis: This traditional method searches for the absence of physical camera hardware characteristics. Specifically, it looks for Photo Response Non-Uniformity (PRNU), a unique sensor noise pattern caused by microscopic silicon defects present in all physical camera sensors. Because AI images are computationally generated and never pass through a physical lens or sensor, they lack PRNU. While theoretically sound, we found this approach highly impractical for our specific use case. When dealing with heavily compressed, web-scraped e-commerce images (which often undergo aggressive JPEG compression), PRNU traces are frequently destroyed even in authentic photos, rendering the analysis highly inaccurate.

Deep Feature Extraction (Our Approach): This method leverages the power of transfer learning. By utilizing robust, pre-trained image recognition networks like ResNet-50 or MobileNetV2, researchers can tap into deep feature representations trained on millions of real-world photographs (such as the ImageNet database). The network learns the "natural" distribution of real-world textures and lighting. When presented with an AI image, the network identifies uncanny anomalies and mathematical regularities in the AI's texture generation. This paradigm strongly aligns with our research goals, as we specifically focus on fine-tuning deeper network layers to expose the hidden mathematical regularities of synthetically generated pixels.

III. SYSTEM CONFIGURATION AND HARDWARE CONSTRAINTS

Processing high-resolution image datasets for deep learning necessitates substantial computational infrastructure. Early in the project, we realized that attempting to train CNNs on local CPU hardware would be unfeasible due to excessive training times and memory bottlenecks.

A. Hardware Infrastructure

To meet our processing requirements, we transitioned to a high-performance cloud computing environment. We utilized a Google Colab Pro instance equipped with a Tesla T4 GPU. The T4 provides 16GB of GDDR6 VRAM. This memory capacity was an absolute necessity; it allowed us to maintain a moderate batch size of 32 while processing our target resolution of 224x224 images. Attempting a batch size of 64 consistently resulted in Out of Memory (OOM) fatal errors, forcing us to carefully balance our training throughput against hardware limitations.

B. Software Stack

We adopted a standardized and highly documented software stack to ensure reproducibility:

- **Core Language:** Python 3.10.
- **Deep Learning Framework:** TensorFlow 2.x was chosen as our primary backend, utilizing the Keras API for its rapid prototyping capabilities and intuitive model summary outputs.
- **Data Manipulation:** Pandas and NumPy were heavily utilized for building our custom data pipelines, handling label matching, and performing array-level image normalizations.
- **Visualization:** Matplotlib and Seaborn were used extensively to chart our loss/accuracy curves and generate confusion matrices to better understand model biases. Scikit-Learn provided the underlying logic for our strict metric evaluations.

IV. DATA INTEGRITY AND PRE-PROCESSING

The old adage "garbage in, garbage out" is acutely true in deep learning. A poorly structured or noisy dataset will actively sabotage even the most mathematically sound neural network architecture.

A. Data Integrity and Path Mapping

We were provided with a raw dataset comprising 4,800 images scraped from Etsy listings. Much of our early, grueling development work was focused on creating a bulletproof data pipeline. The images were delivered in a bulk ZIP archive, while the corresponding ground-truth labels were isolated in a separate CSV file. This detached structure presented a high risk for data mismatching during the loading phase.

To mitigate this, we engineered a custom Python script utilizing the 'os' and 'pandas' libraries. This script programmatically verified the existence of every single image file referenced in the CSV, flagged and removed any corrupted

or zero-byte files, and strictly mapped the valid file names to their respective labels.

Furthermore, class imbalance can cause a network to lazily predict the majority class to minimize loss. We audited our mapped data to ensure a relatively balanced distribution, resulting in a finalized, clean dataset of 2,485 original (Authentic) images and 2,315 AI-generated (Synthetic) images.

B. The Resolution Gap Challenge

Perhaps the most pivotal turning point in our research was identifying the "Resolution Gap." Our initial baseline tests utilized images aggressively downsampled to 32x32 pixels to save training time. At this microscopic scale, a photograph of a genuine, hand-stitched leather handbag and a diffusion-generated image of a handbag are mathematically almost identical; the compression destroys the very artifacts we need to detect.

By electing to use 224x224 as our standardized input shape, we effectively fed the model 49 times more spatial information per image. While computationally expensive, this resolution allows the CNN's filters to detect micro-artifacts: illogical shifts in fabric weaves, mathematically "too perfect" lighting gradients, or structural impossibilities in metallic reflections. These are the tell-tale signs of current GenAI.

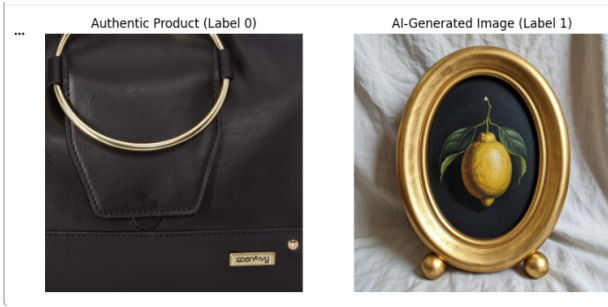


Fig. 1. Visual context: While authentic items (left) exhibit natural imperfections in texture, AI generations (right) often display an uncanny, mathematically smooth surface devoid of true physical wear.

V. METHODOLOGY

A. Why MobileNetV2?

When selecting a base model for transfer learning, we evaluated several architectures including ResNet-50 and VGG16. We ultimately selected MobileNetV2 due to its elegant use of an inverted residual structure and Depthwise Separable Convolutions.

Standard convolutions perform channel-wise and spatial-wise computations in a single step, which is highly parameter-intensive. Depthwise Separable Convolutions split this into two distinct layers: a depthwise convolution for spatial filtering, followed by a 1x1 pointwise convolution for channel combination. This drastic reduction in parameters allowed us to train deeper networks much faster on our Colab instance while preserving a sufficiently large receptive field to capture complex textural anomalies.

TABLE I
DETAILED ARCHITECTURAL HEAD CONFIGURATION

Layer Type	Output Shape	Activation
MobileNetV2 Base	(7, 7, 1280)	ReLU6
GlobalAvgPooling2D	(1280)	—
Dense (512 Units)	(512)	ReLU
Dropout (0.5 Rate)	(512)	—
Output Dense	(1)	Sigmoid

B. Fine-Tuning Strategy: The "Goldilocks" Zone

Training a deep network from random weights on a dataset of only 4,800 images would almost certainly result in severe overfitting. Therefore, we relied heavily on transfer learning using weights pre-trained on ImageNet.

However, simply training the top classification head was insufficient; the model needed to learn the specific, novel artifacts of AI generation. We adopted a partial unfreezing strategy. We froze the first 114 layers (which handle basic feature extraction like edges and simple curves) and unfroze the top 40 convolutional layers.

This required careful trial and error. Freezing too many layers left the model too generic to detect AI nuances. Conversely, unfreezing the entire network caused "Catastrophic Forgetting," where the model destroyed its pre-learned ImageNet weights and wildly overfit to our small training set. The final 40 layers proved to be the "Goldilocks zone" for processing complex, composite textures.

C. Regularization and Loss Formulation

To further combat the persistent threat of overfitting, we implemented aggressive regularization. We applied a 50% Dropout layer immediately preceding the final classification node.

For our objective function, we utilized standard Binary Cross-Entropy, given the two-class nature of our problem:

$$L = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

Where y_i is the true label (0 or 1) and $\hat{y}_i$ is the predicted probability. We paired this with the Adam optimizer, utilizing a low learning rate of 0.0001 to ensure stable weight updates without overshooting the local minima during the fine-tuning of the un-frozen layers.

VI. EXPERIMENTS AND TUNING

A. Training Dynamics and Overfitting Management

Our final training run was orchestrated over 12 epochs. We closely monitored the divergence between training and validation metrics. Around Epoch 6, we noticed a distinct behavioral shift. The training accuracy began climbing aggressively toward 98%, while the validation accuracy plateaued and hovered around 72.7%.

This divergence is a classic hallmark of a model beginning to memorize the training data rather than learning generalized

features. We faced a choice: implement complex data augmentation to artificially expand the dataset, or halt training to prevent further divergence and optimize the decision logic post-training. Given the specific nature of AI artifacts (where standard augmentations like blurring or color-shifting might accidentally mimic or destroy the very artifacts we were trying to detect), we opted to halt training and focus on Decision Optimization.

B. The Threshold Grid-Search Calibration

Standard binary classifiers default to a 0.5 decision threshold—if the model outputs a probability higher than 50%, it predicts Class 1. This assumes the model’s confidence is perfectly calibrated.

Upon analyzing our confusion matrix, we realized our model was heavily skewed; it was highly aggressive in predicting Label 1 (Synthetic). It was successfully catching the AI images, but at the cost of flagging perfectly valid, high-quality authentic images as fake.

To evaluate this, we looked at the F1-score, which provides the harmonic mean of Precision and Recall:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

To fix the model’s aggressive bias, we performed a manual grid search, sliding the classification threshold incrementally from 0.10 to 0.85. We found that by pushing the threshold up to 0.65, we forced the model to be more “certain” before branding an image as AI. This successfully filtered out the bulk of our “False Positives” (clean, real photos) without significantly damaging our ability to catch the actual fakes. This single post-processing calibration was the catalyst for achieving our peak F1-score of 0.7719.

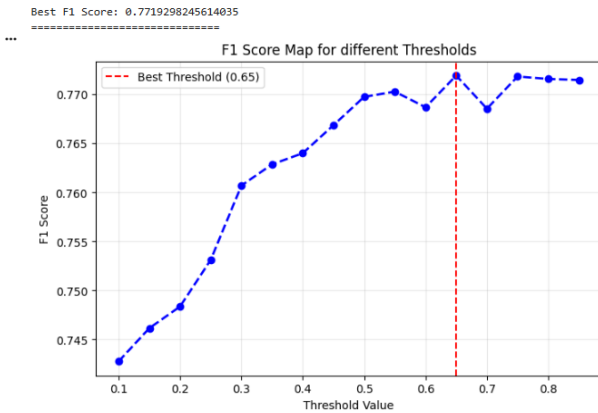


Fig. 2. Threshold Grid-Search Analysis: We observed a distinct peak in the F1-score when the decision threshold was elevated to 0.65, effectively balancing Precision and Recall.

VII. DISCUSSION AND QUALITATIVE ANALYSIS

Metrics only tell a partial story. To truly understand our model’s behavior, we conducted a qualitative visual inspection of the failure cases—the instances where the model confidently made the wrong decision.

A. Anatomy of a False Positive

A False Positive occurs when an authentic image is incorrectly flagged as AI. We noticed a distinct pattern here: the misclassified real images were almost exclusively high-end, professionally shot studio photographs. These images featured perfect lighting, completely isolated white backgrounds, and deep depth-of-field blurring. Ironically, because generative AI is heavily trained on professional studio photography, a “perfect” real photograph mathematically resembles an AI generation to the neural network. The model is effectively punishing authentic sellers for having overly polished marketing material.

B. Anatomy of a False Negative

Conversely, a False Negative occurs when an AI-generated image slips past the detector. These were predominantly images that were intentionally low quality, contained added noise, or simulated vintage, blurry cameras. Generative AI usually defaults to a baseline of “sharp smoothness.” When an AI image deviates from this and incorporates heavy noise or blur, our model’s texture-focused filters failed to trigger, assuming the imperfections were proof of a physical camera.

VIII. ETHICAL CONSIDERATIONS

Deploying automated AI detection systems in a live e-commerce environment carries significant ethical weight. A False Positive in an academic setting is merely a statistical error; on Etsy, it could result in the automatic suspension of a legitimate artisan’s storefront, directly impacting their livelihood.

This reality heavily influenced our decision to move the threshold to 0.65. In a platform moderation context, we argue that it is better to let a few synthetic images slip through (prioritizing high Precision) than to aggressively ban innocent sellers (lowering Recall for authentic items). AI detection should serve as an initial flagging mechanism for human trust-and-safety teams, rather than a fully autonomous judge and jury.

IX. FUTURE WORK

While our fine-tuned MobileNetV2 architecture proved successful, the generative AI landscape is moving too fast for static solutions. Future iterations of this research should explore Vision Transformers (ViTs). Unlike CNNs, which look at localized pixel clusters, Transformers analyze an image globally using self-attention mechanisms. This would allow a model to understand that while the texture of a generated hand might look mathematically fine on a micro-level, the fact that it has six fingers on a macro-level dictates that it is synthetic.

Additionally, exploring ensemble methods—where a lightweight CNN evaluates textures while a separate model evaluates frequency domains—could provide a more robust, multi-layered defense against future generative architectures.

X. CONCLUSION

In this study, Ajay Kumar Singh and Prashant Kumar Sengar have demonstrated that despite the astonishing visual quality of modern Generative AI, these systems still leave behind mathematically detectable traces. By discarding the limitations of low-resolution inputs and embracing a 224x224 spatial analysis, coupled with the strategic 40-layer fine-tuning of a MobileNetV2 backbone, we successfully isolated these synthetic anomalies. Furthermore, by manually calibrating our decision boundary to a 0.65 threshold, we mitigated the model's inherent bias, protecting legitimate vendors while identifying fraudulent listings. While generative models will continue to improve, our findings provide a scalable, highly effective defensive framework to help preserve the foundational authenticity and trust of the Etsy marketplace.

MEMBER CONTRIBUTIONS

Ajay Kumar Singh: Engineered the initial data loading pipelines, managed the complex image path mapping logic, constructed the Keras model architecture, and oversaw the GPU training iterations, including hardware memory management.

Prashant Kumar Sengar: Executed the threshold grid-search calibrations, designed the data visualization and F1-score graphing, conducted the qualitative analysis of failure cases, and led the final technical reporting and documentation.

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